

ISE 5970: Advanced Systems Optimization

University of Oklahoma School of Industrial and Systems Engineering

Spring 2025

Instructions for Assignment Submission:

1. The assignment must be uploaded to Canvas by the indicated due date. The solution should be uploaded to Canvas as a PDF file.
2. Along with your main submission (in PDF format), please ensure you also attach any supplementary files (Excel, Python) that were instrumental in your work. If any Python code was used, upload it in HTML format.

Assignment 2: Dual problem, Gommory cuts and Column generation

1. (25 points) Multi-Commodity Network Flow Problem: Primal-Dual example

A logistics company is managing the distribution of five different products through a network of 8 nodes. The nodes represent distribution centers and retail outlets. Shipments along each bi-directed arc incur a cost that depends on the product (commodity) being shipped. Each arc has a capacity constraint that limits the total number of units (across all commodities) that can be transported along that arc. The company wishes to minimize the total shipping cost while satisfying supply and demand for each product and not exceeding arc capacities. You can create a unmet demand parameter if need it, just add a big enough cost to it, like 100.

Graph components

- **Nodes:** $N = \{1, 2, 3, 4, 5, 6, 7, 8\}$
- **Arcs:** $A = \{(1, 2), (1, 3), (2, 4), (2, 5), (3, 6), (3, 7), (4, 8), (5, 8), (6, 8), (7, 8)\}$

There are five products (commodities), indexed by $k \in \{1, 2, 3, 4, 5\}$. Their supply and demand are given in the following table:

Commodity	Source Node	Sink Node	Demand (units)
1	1	4	10
2	1	5	8
3	2	6	6
4	3	7	7
5	1	8	5

Arc Capacities

For each arc $(i, j) \in A$, the capacity u_{ij} (maximum total units across all commodities) is specified as:

$$\begin{aligned}u_{1,2} &= 15, & u_{1,3} &= 15, \\u_{2,4} &= 10, & u_{2,5} &= 10, \\u_{3,6} &= 10, & u_{3,7} &= 10, \\u_{4,8} &= 12, & u_{5,8} &= 12, \\u_{6,8} &= 12, & u_{7,8} &= 12.\end{aligned}$$

Cost Coefficients Let c_{ij}^k denote the shipping cost per unit for commodity k along arc (i, j) . The following

table gives illustrative costs:

Arc	c_{ij}^1	c_{ij}^2	c_{ij}^3	c_{ij}^4	c_{ij}^5
(1, 2)	2	3	2	3	2
(1, 3)	3	2	3	2	3
(2, 4)	1	2	1	2	1
(2, 5)	2	1	2	1	2
(3, 6)	1	2	1	2	1
(3, 7)	2	1	2	1	2
(4, 8)	3	3	3	3	3
(5, 8)	2	2	2	2	2
(6, 8)	1	1	1	1	1
(7, 8)	2	2	2	2	2

- (5 points) Formulate the multi-commodity network flow problem.
 - (5 points) Formulate the dual problem associated to the primal network flow problem.
 - (4 points) Explain the relationship between the dual variables and the reduced cost of the primal.
 - (4 points) Prove that the dual objective function for a primal minimization problem is always greater than (or equal to) the objective function of any feasible dual solution.
 - (4 points) Prove that at optimality, if the primal minimization problem has a bounded feasible solution then there exists a feasible solution to the dual problem.
 - (3 points) Explain the concept of complementary slackness.
2. (50 points + 10 extra points) **Multi-Commodity Network Flow Problem: Column generation**
Following the same context as the network flow model from the previous homework, imagine that you are focusing on one corridor in the transportation network. In this corridor, there are 8 stations. The tables below provide the OD demand and transportation times between each pair of stations. Use these data to design routes and assign buses as described in the following section.

OD Demand Matrix (in number of passengers)

OD	1	2	3	4	5	6	7	8
1	0	150	300	200	100	250	180	220
2	160	0	210	270	130	190	240	150
3	240	180	0	310	140	260	170	200
4	200	220	190	0	280	210	320	160
5	130	140	150	160	0	230	250	290
6	210	170	260	230	310	0	180	150
7	190	240	200	270	220	160	0	210
8	250	200	150	180	210	230	190	0

Transportation Times Matrix (in minutes)

OD	1	2	3	4	5	6	7	8
1	0	10	15	20	25	30	35	40
2	12	0	18	24	16	20	28	32
3	14	16	0	22	26	30	18	24
4	20	24	26	0	12	18	22	28
5	18	14	20	16	0	15	25	30
6	25	20	22	28	30	0	14	19
7	30	28	24	20	18	15	0	17
8	35	32	28	24	20	18	16	0

Route Construction and Bus Allocation:

A **route** is defined as a sequence of arcs (direct connections between stations). For this exercise:

- **Directionality:** Routes must consider only combinations in a single direction. For example, if a route includes the arc (Node 1, Node 2) followed by (Node 2, Node 5), then the route cannot include any arc that reverses direction (e.g., (Node 5, Node 3)). This ensures that the routes are consistently oriented and do not backtrack.
- **Cost Structure:** Using a route incurs a **fixed cost of \$1000** per route, plus an **additional cost of \$100 per station** visited. For example:
 - A route visiting two stations incurs a cost of \$1200.
 - A route visiting three stations incurs a cost of \$1300.
- **Bus Assignment:** Each route must be served by buses. Each bus has a capacity of 150 passengers, and you must determine how many buses must be assigned to each route to satisfy the demand for passengers.
- **Bus Availability Constraints:** No more than 5 buses can be assigned to a single route.
- **Total time for the transportation system:** To have more interesting results, add a constraint that limits the aggregate total time for all passengers in the system to be less or equal to: 391890.

- (10 points) Formulate the mixed integer linear program for the route problem.
- (10 points) Solve the problem using Gurobi (or any optimizer of your preference) and report the following:
 - The solution routes.
 - The number of buses assigned to each route.
 - The objective function value.
 - The computational time required to reach the optimum.
- (5 points) Explain, in your own words, why this problem is suitable for implementing the column generation technique.
- (15 points) Create the reformulation for the problem, clearly delineating:
 - The master problem.
 - The auxiliary (or pricing) problem.
- (10 points) Solve the new reformulation and report the following:
 - The solution routes.
 - The number of buses assigned to each route.
 - The objective function value.
 - The computational time required to reach the optimum.
- (10 extra points) Using the network that was provided for the previous homework (homework 1, Traffic flow optimization problem: Designing routes and allocating buses), use the column generation technique to solve the route problem and report your findings.
 - The solution routes per corridor.
 - The number of buses assigned to each route.
 - The objective function value.
 - The computational time required to reach the optimum.

3. (25 points) **Integer programming: Branch and bound and cutting plane techniques**

A manufacturing company produces three products: **Prod1**, **Prod2**, and **Prod3**. The company aims to determine the optimal production levels for these products to maximize its total profit. However, production decisions are constrained by limited processing capacity and market requirements. Specifically, there is a cap on the total available processing time, as well as upper and lower bounds on the number of units that can be produced for each product. The objective is to maximize the total profit generated from the production of the three products. The following parameters define the problem:

Product	Profit per Unit	Processing Time	Maximum Demand	Minimum Production
Prod1	20	4.6	102.3	15.3
Prod2	40	7.5	357.4	23.2
Prod3	32	2.4	182.6	17.5

Table 1: Summary of Coefficients per Product

The total available processing time is 556.7 hours. The maximum demand specifies the upper limit on the number of units that can be produced for each product, whereas the minimum production level indicates the least amount that must be produced.

- (a) (5 points) Solve the LP relaxation of the problem (i.e. ignore the integrality constraints). Report:
 - The optimal solution (x_1^*, x_2^*) of the LP relaxation.
 - The optimal value of the objective function z^* .
- (b) (5 points) Determine whether the LP relaxation solution is integer. If not, generate the branch and bound tree until reach optimality, please explain your reasoning at each pruned branch.
- (c) (5 points) Explain the search difference when doing depth first, best node first and least-discrepancy.
- (d) (5 points) A common technique to speed up the process of finding the optimal solution is to include in the process a cutting plane inequality, explain the intuition behind the cuts and how both techniques can be used together.
- (e) (5 points) Now, repeat the branch and bound process but before branching and after solving the linear relaxation, create a cut and add it to the problem. Solve again the problem and show the branch and bound tree, each cut added and all the calculations or reasoning behind each step.